

Maru Sathvik Reddy

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SKILLS

- Languages:** Python, C++
 - AI/ML Libraries:** PyTorch, Transformers, Scikit-learn, NumPy, Pandas, OpenCV, LangChain, LangGraph
 - Tools/Platforms:** MySQL, Git, GitHub, VS Code, Docker, Reids
 - Soft Skills:** Problem-Solving, Team Work, Adaptability

PROJECTS

- Multi-Agent Research Assistant:** | [GitHub](#)

Mar' 26

 - Engineered a fully local multi-agent research pipeline using LangGraph with Supervisor, Planner, Researcher, Analyst, and Writer agents collaborating via a stateful routing loop, async parallel web scraping, and an in-flight lock preventing duplicate concurrent graph executions.
 - Integrated a 3-layer Redis Stack caching architecture with prompt-level LLM cache, versioned report storage via Redis Lists, and HNSW vector similarity search using all-MiniLM-L6-v2 embeddings for semantic cache hits enabling hybrid report generation merging cached context with fresh scraped insights.
 - Developed an agentic chat interface with intent-based routing to live web search and URL scraping tools, session-persisted context accumulation, and a Dockerized stack with Nginx, MemorySaver graph checkpointing, and SHA256-derived thread IDs for crash recovery.

Tech: Python, LangGraph, LangChain, Ollama, FastAPI, Redis, Docker, Nginx, Streamlit
- Multi-Modal Retrieval Augmented Generation:** | [GitHub](#)

Mar' 26

 - Designed a multi-modal RAG architecture enabling document, image, and timestamp-aware video ingestion with context-grounded generation using LLaMA3 (Ollama) and quantized LLaVA vision models.
 - Built a scalable FastAPI backend with background ingestion pipelines, semantic chunking, batched GPU embeddings, ChromaDB vector indexing, and streaming token-level LLM responses to minimize hallucination and latency.
 - Developed an interactive Streamlit-based conversational interface integrating controlled ingestion workflows, real-time token streaming, and multimodal semantic retrieval, optimizing inference through model quantization, frame-differencing, and efficient vector indexing strategies.

Tech: Python, PyTorch, Transformers, LangChain, FastAPI, ChromaDB, Sentence-Transformers, OpenCV, Ollama, Streamlit, Docker
- Autoregressive Transformer for Python Code Generation:** | [GitHub](#)

Nov' 25

 - Implemented a decoder-only Transformer for autoregressive Python code generation using a Python subset of the CodeParrot dataset with GPT-style tokenization
 - Constructed causal self-attention, positional embeddings, token embeddings, and next-token prediction for code completion tasks.
 - Established a training and inference pipeline supporting prompt-based generation with greedy decoding

Tech: Python, PyTorch, Transformer Decoder, Self-Attention, GPT Tokenization, NLP

TRAINING

- Machine Learning - Elevate Labs**

Apr' 25 – May' 25

 - Gained hands-on experience in Supervised and Unsupervised Machine Learning, with a strong understanding of core algorithms, feature engineering, and model evaluation techniques.
 - Built a resume ranking system using cosine similarity to match resumes with job descriptions, automating candidate screening through text preprocessing and relevance scoring.

Tech: Python, Pandas, Scikit-learn, NLP

CERTIFICATES

- Generative AI, LLM & RAG | [GeeksforGeeks](#) Feb' 26
 - Foundational ML concepts | [AWS ML Exam Basics](#) Oct' 25
 - A Guide to Machine Learning with Data Science | [Cipher Schools](#) Jul' 25
 - Deep Learning with TensorFlow | [IBM Cognitive Class](#) Mar' 25

ACHIEVEMENT

- Achieved Global Rank 588 (Top 15%) in Kaggle's "Predicting Road Accident Risk" competition Nov' 25

EDUCATION

- Lovely Professional University**

Punjab, India
Bachelor of Technology - Computer Science and Engineering; CGPA: 7.28
Aug' 23 - Present
- Sri Chaitanya Junior College**

Hyderabad, Telangana
Intermediate; Percentage: 88.9
Jul' 21 - Mar' 23
- Wisdom Spaes School**

Metpally, Telangana
Matriculation; Percentage: 100
Jul' 20 - Mar' 21